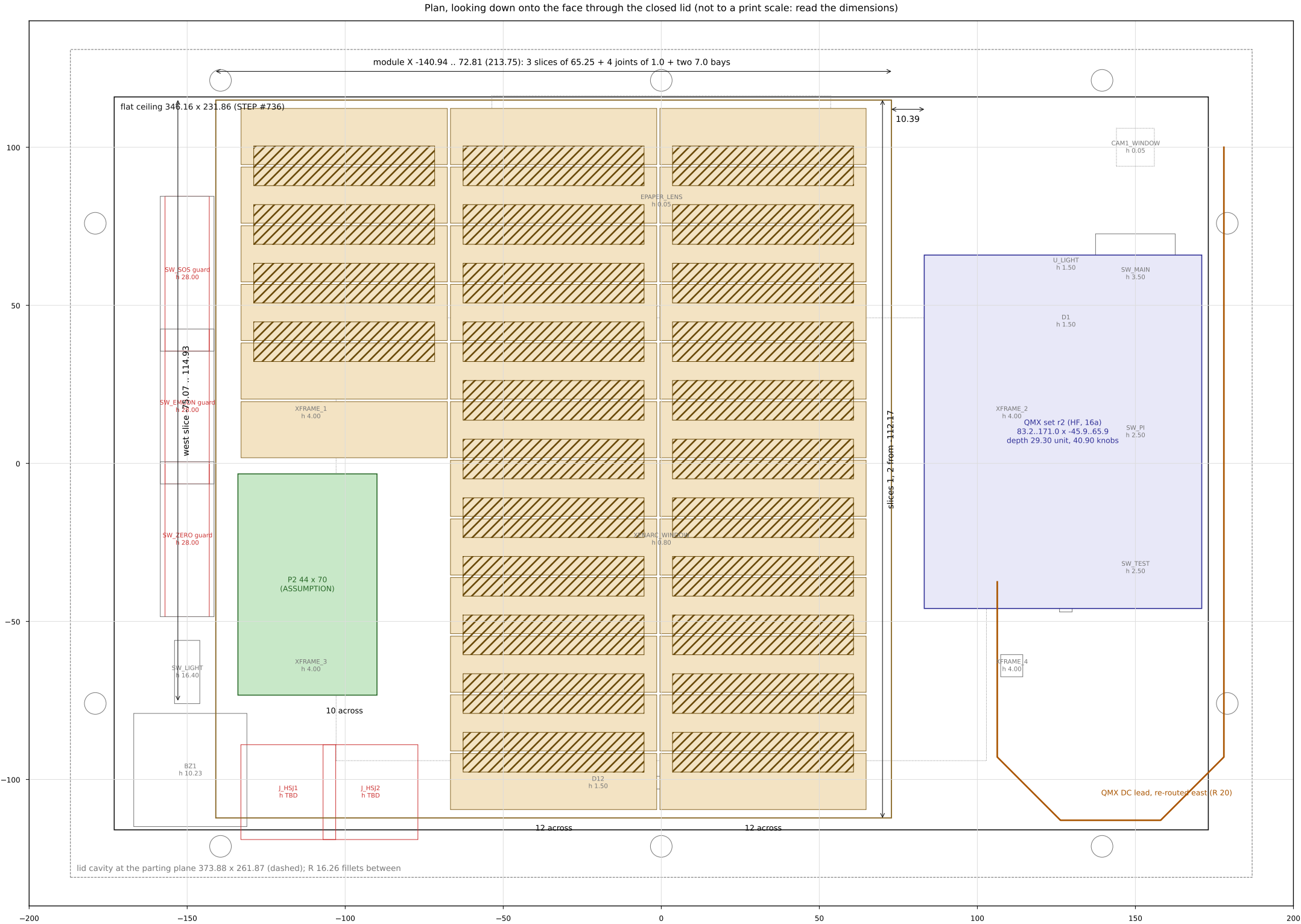


MeshSat field kit V2, Option A(i) lid pack: sheet A1-1 arrangement B in plan: the lid module, the QMX set and every face part under the lid

MESHsat-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



ARRANGEMENT B (HF kept, no tablet in the lid)
cells 56: layer 1 30, layer 2 26; 4514P (48 used as instructed, 8 spare places)

CELL Samsung INR18650-35E: d 18.55 max, L 65.25 max, 50 g max
(spec Ver. 1.1, 3.10 and drawing)
BLOCK A06's construction (the ruled base block): cells touching,
wrap 0.5 a side, joints 1.0 at each cell end along X
(nickel strip and fish paper), layer 2 nested in the grooves
HOLD the block bonded to the lid plate (cell glue and a fillet,
adhesive TBD); the cover on 8 M3 standoffs in the end bays
presses a PORON 4701-30 pad 1.0; the plate bonded to the
lid's ceiling with DP8005 as the r2 set (no hole in the case)
VENT every cell end faces its 1.0 joint; the end bays are open
slots; the case relieves through Peli's pressure valve

STACK FROM THE CEILING	nominal
DP8005 bond	0.20
5052-H32 lid plate	2.00
block, one layer / two layers	19.55 / 35.61
PORON pad (compressed)	1.00
cover 1.0 + insulation 0.25	1.25
module depth, one / two layers	24.00 / 40.06
P2 on 2.5 standoffs (ASSUMPTION)	22.47

ROOM face top to ceiling: 47.92 nominal, 44.39 worst,
42.11 sensitivity (frame_seat.out, r2 record B)

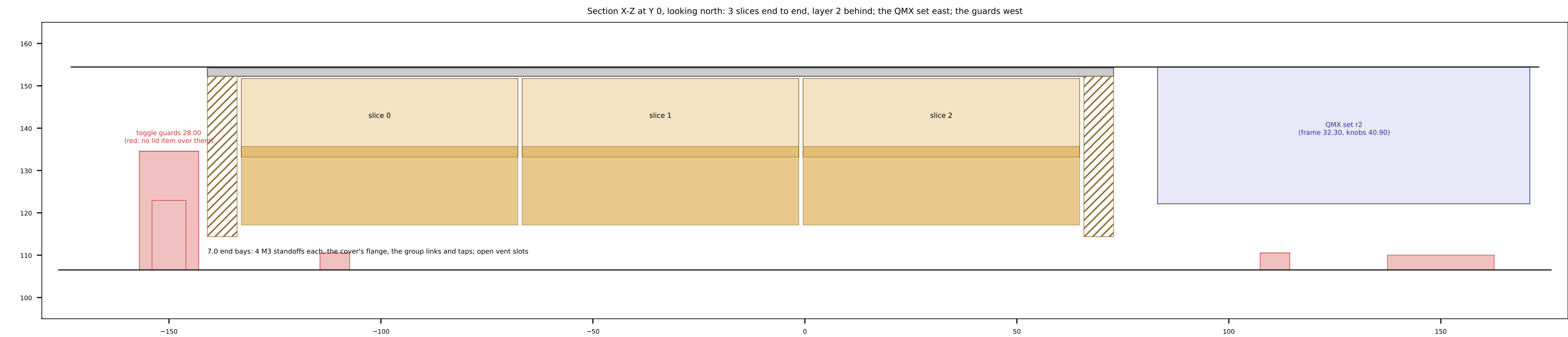
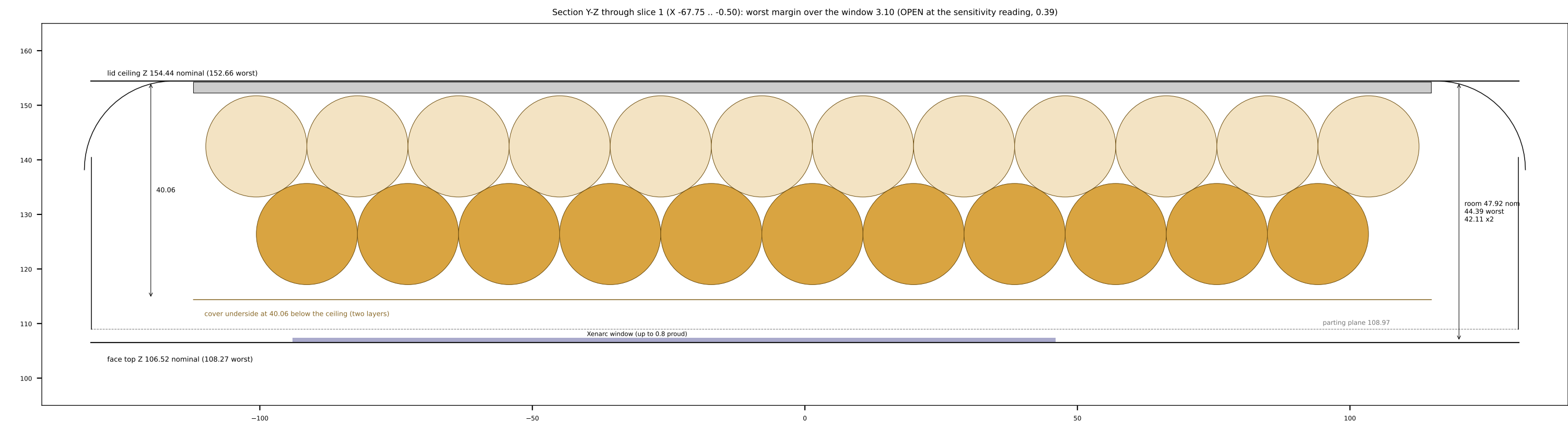
TIGHTTEST ROWS (worst, x2; min 1.0)
two layers over the Xenarc window 0.80: 3.10, 0.39 OPEN
two layers over the LED bar 1.50: 2.40, -0.31 OPEN
two layers over the e-paper lens: 3.85, 1.14 MET
one layer over the XFRAME screws: 15.96, 13.25 MET
layer 2 refused over XFRAME_1 and _3 (4.00): -0.10 worst

PLAN module to the QMX tray 10.39; inside the flat ceiling
1.00 in Y (the fillets begin at +-115.93)
west edge 1.0 beyond the guard caps and the allowance
west slice stops north of the headset jacks (TBD)

OPEN AT THE MOCK-UP: T9 (chalk on the module's cover and the
face parts, lid closed), T8 (pull test of a bonded plate on
Peli's polypropylene), T-A1-1 to T-A1-4 of the README.

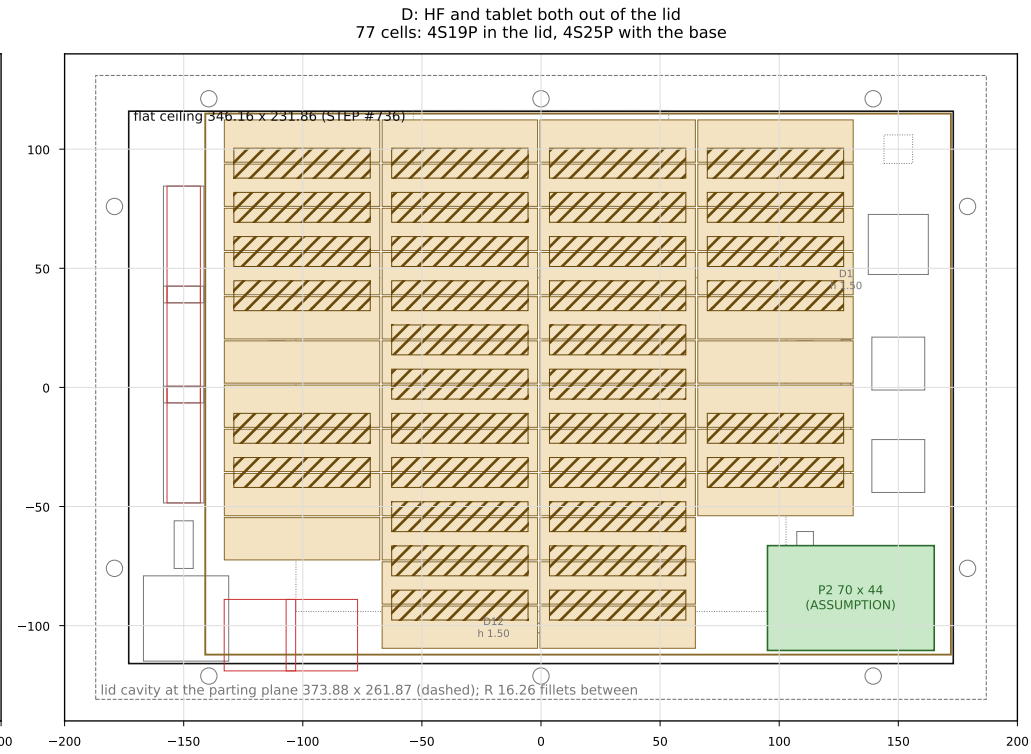
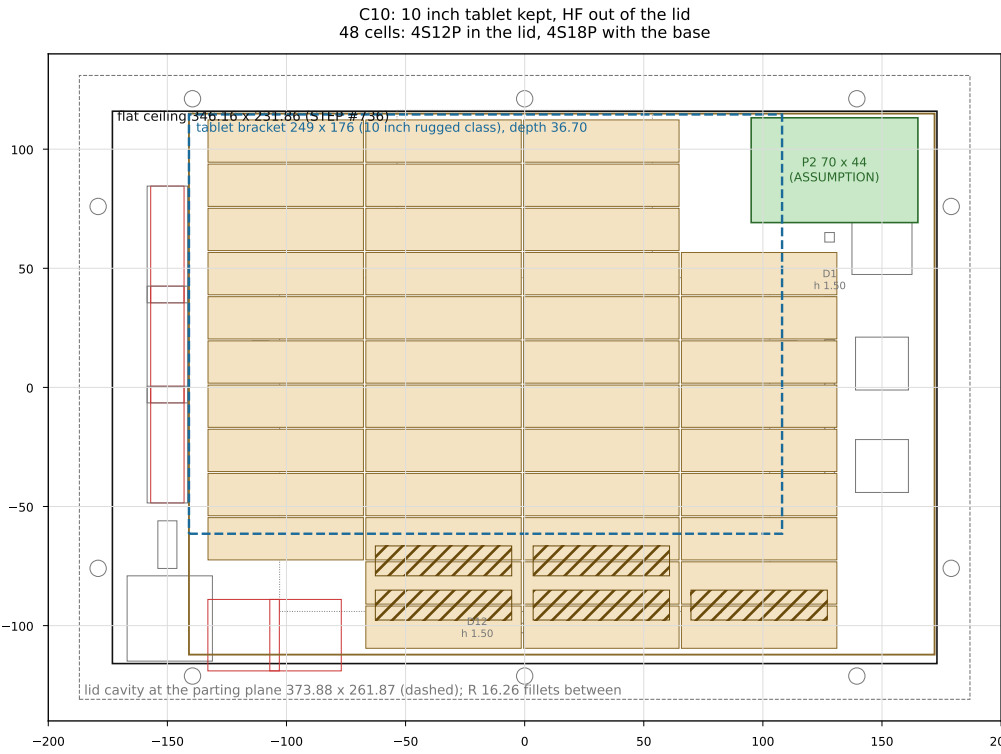
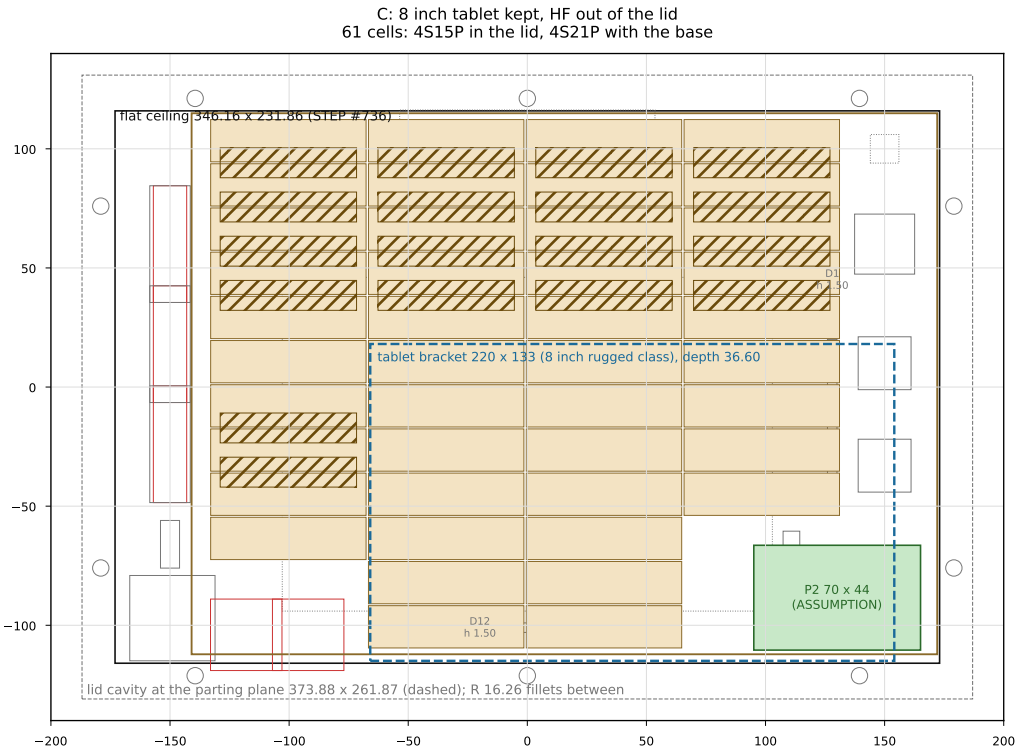
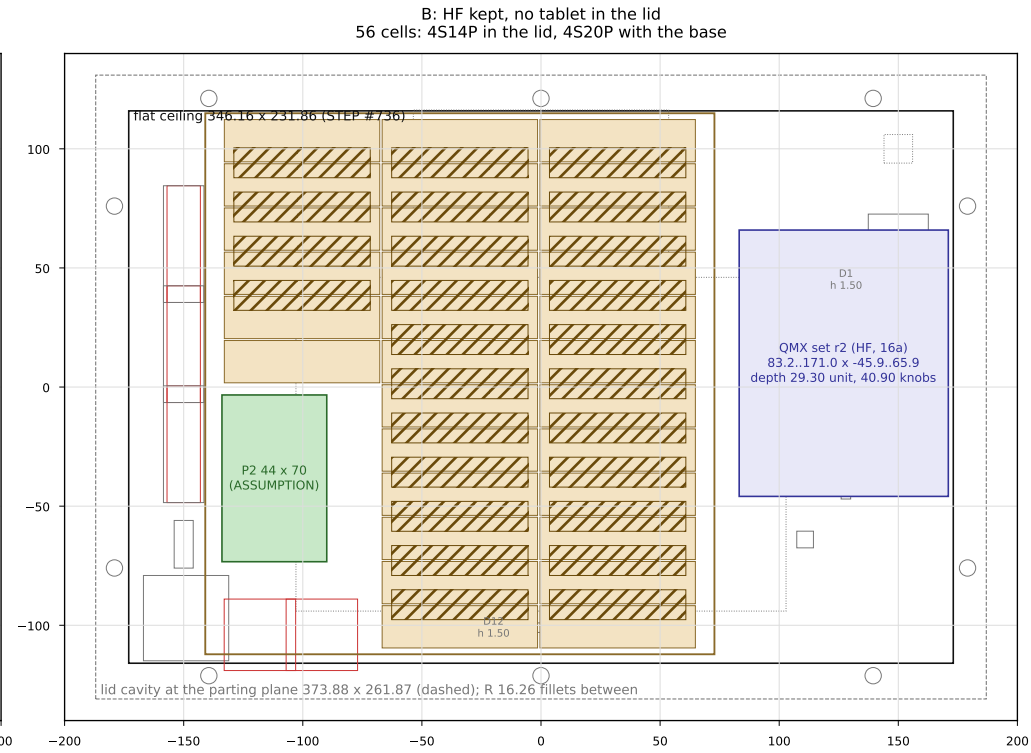
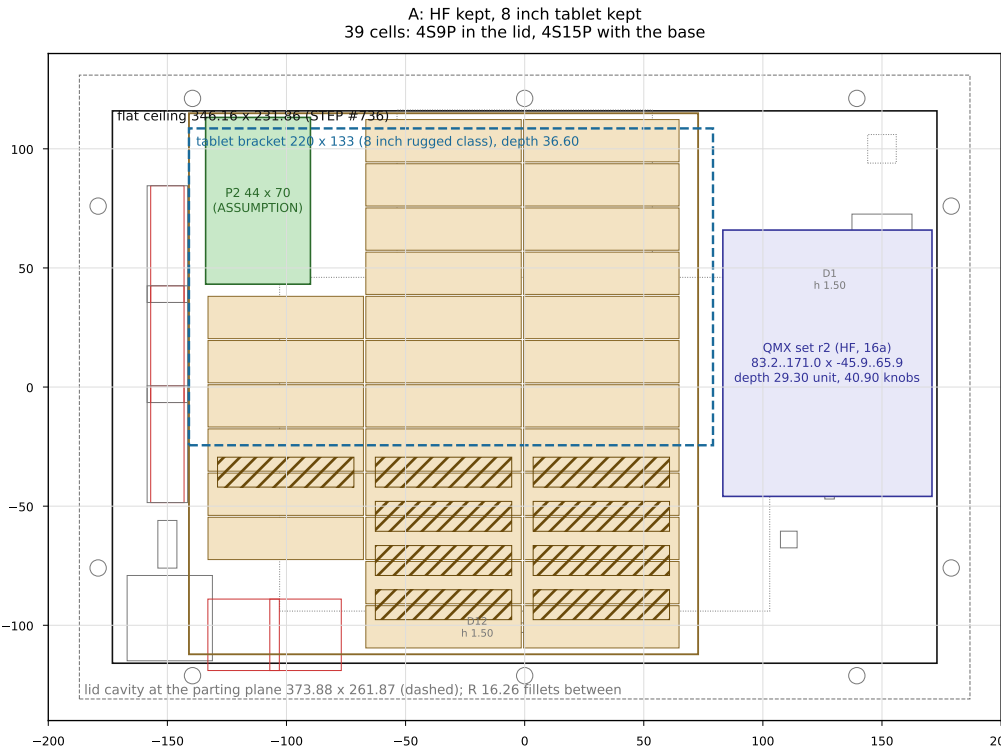
MeshSat field kit V2, Option A(i) lid pack: sheet A1-2 sections: the module under the ceiling over the face, at the worst room

MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



MeshSat field kit V2, Option A(i) lid pack: sheet A1-3 the arrangements that fit, and the owner decision

MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



THE OWNER DECISION (the session does not take it:
it removes an owner-approved function, 6 Sep 2026)

48 cells (4S12P) in the lid with BOTH the QMX HF set (16a) and the tablet bracket (16d): DOES NOT FIT. At the worst room the lid holds 39 places with both, 4S9P, 4S15P in all (the check's B1).

What fits 4S12P (each with every row of sheet A1-1 at or above its minimum at the worst):

1. HF kept, tablet bracket out of the lid: 56 places (4S14P); the tablet goes out of the case.
2. 8 inch tablet kept, HF (QMX) out of the lid: 61 places (4S15P); HF inside (16a) is lost: the base has no volume for the QMX set
3. Both kept, lid pack 4S9P (4S15P in all): not Option A(i). The coordinator's two-pack run (Sept. reference day, 200 W stage): lid at its 13.23 C basis NOT MET at 400, 650, 1000 Wp; lid at +20 C NOT MET at 400 Wp, met at 650 (2.0 Wh left) and 1000 Wp (29.4 Wh left)

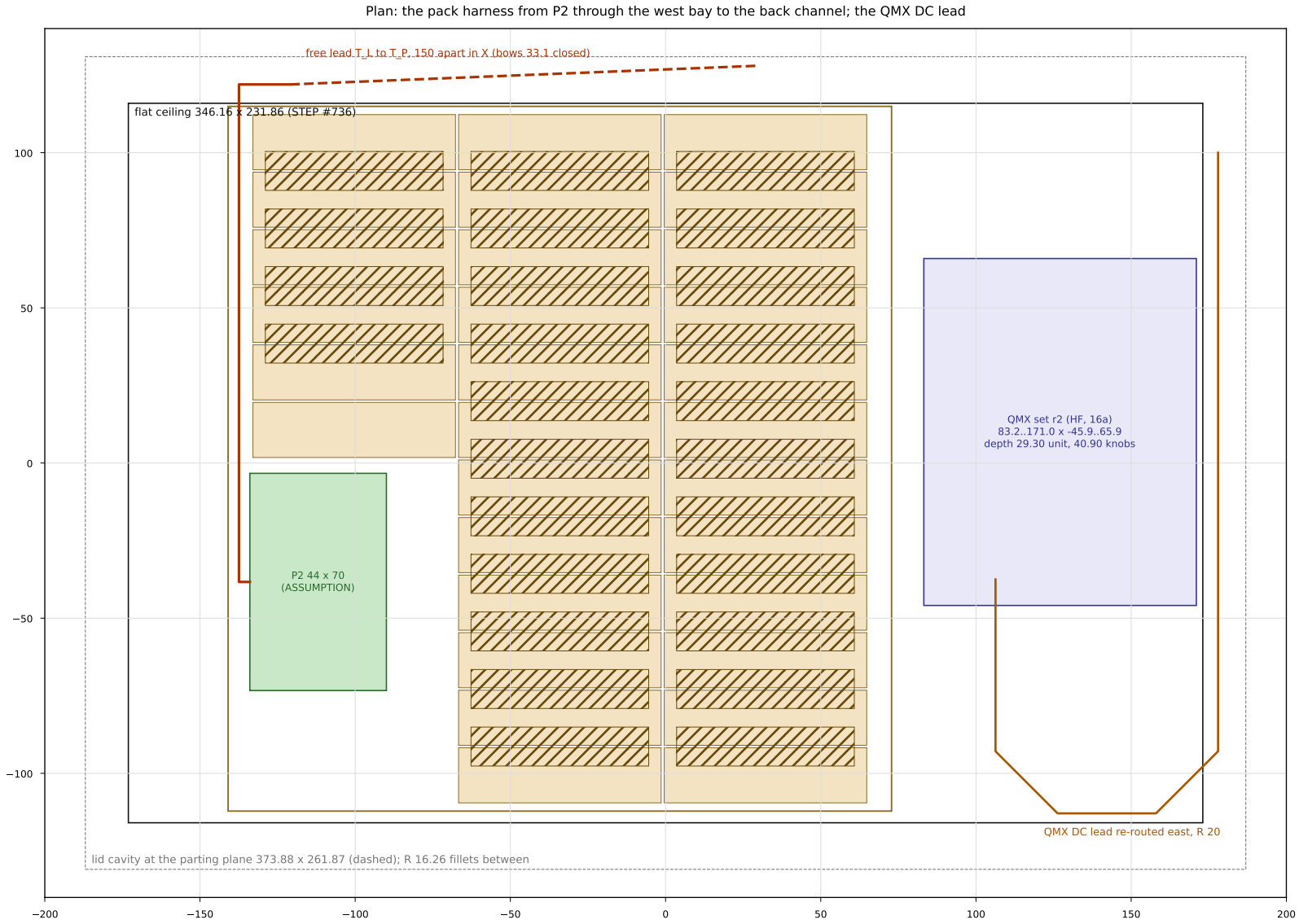
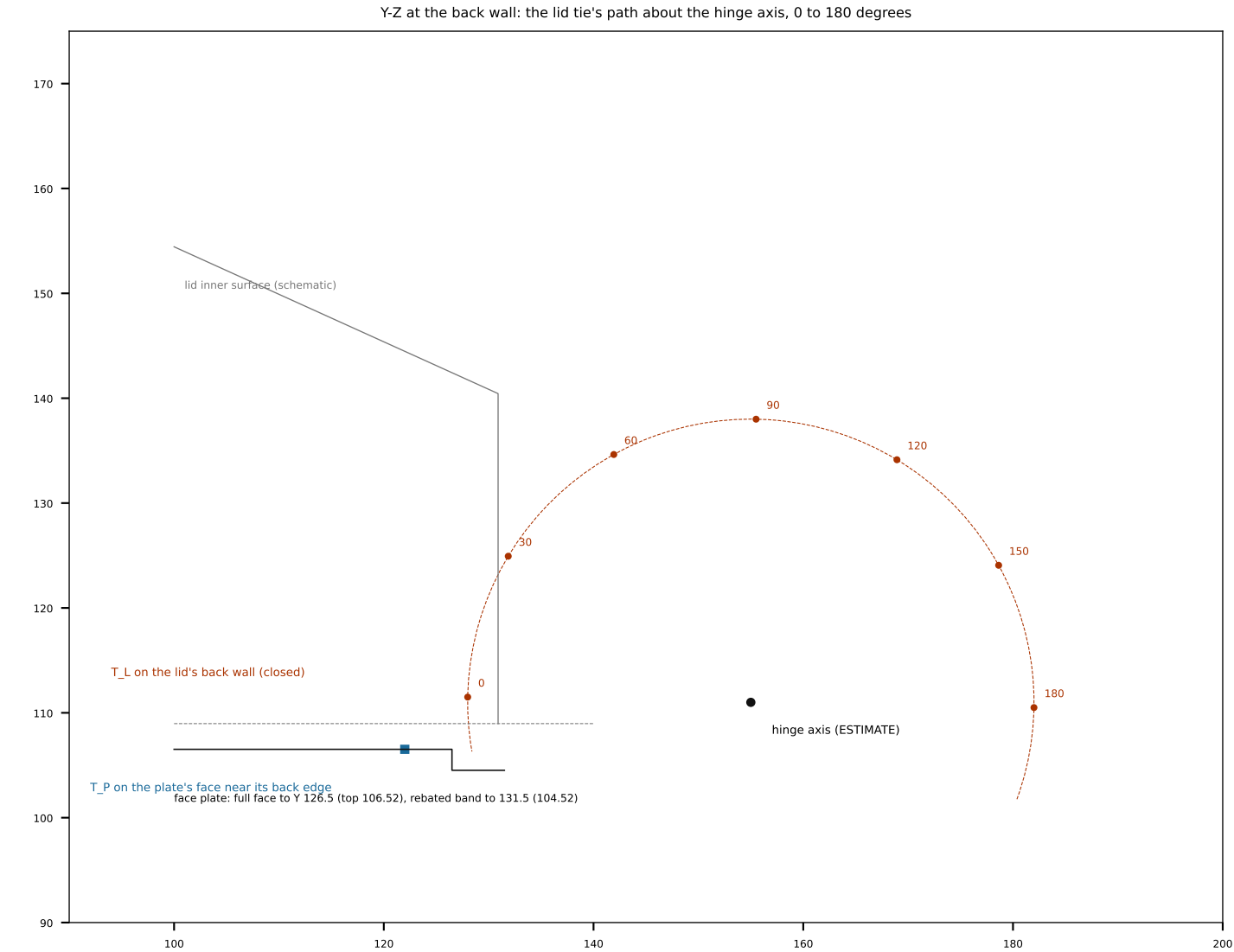
10 inch tablet: only with HF out: 48 places (4S12P). Both out: 77 places (4S19P).

Session's recommendation (engineering view only): option 1, which keeps a radio bearer and gives 8 spare places; the tablet works from outside the case on the kit's WiFi and USB-C outlet.

D-01 already defers HF, the tablet bracket and a second pack from prototype 1: Option A(i) reopens D-01 whichever option is taken.

MeshSat field kit V2, Option A(i) lid pack: sheet A1-4 the hinge harness: route, travel, ties, disconnect; the QMX DC lead re-routed

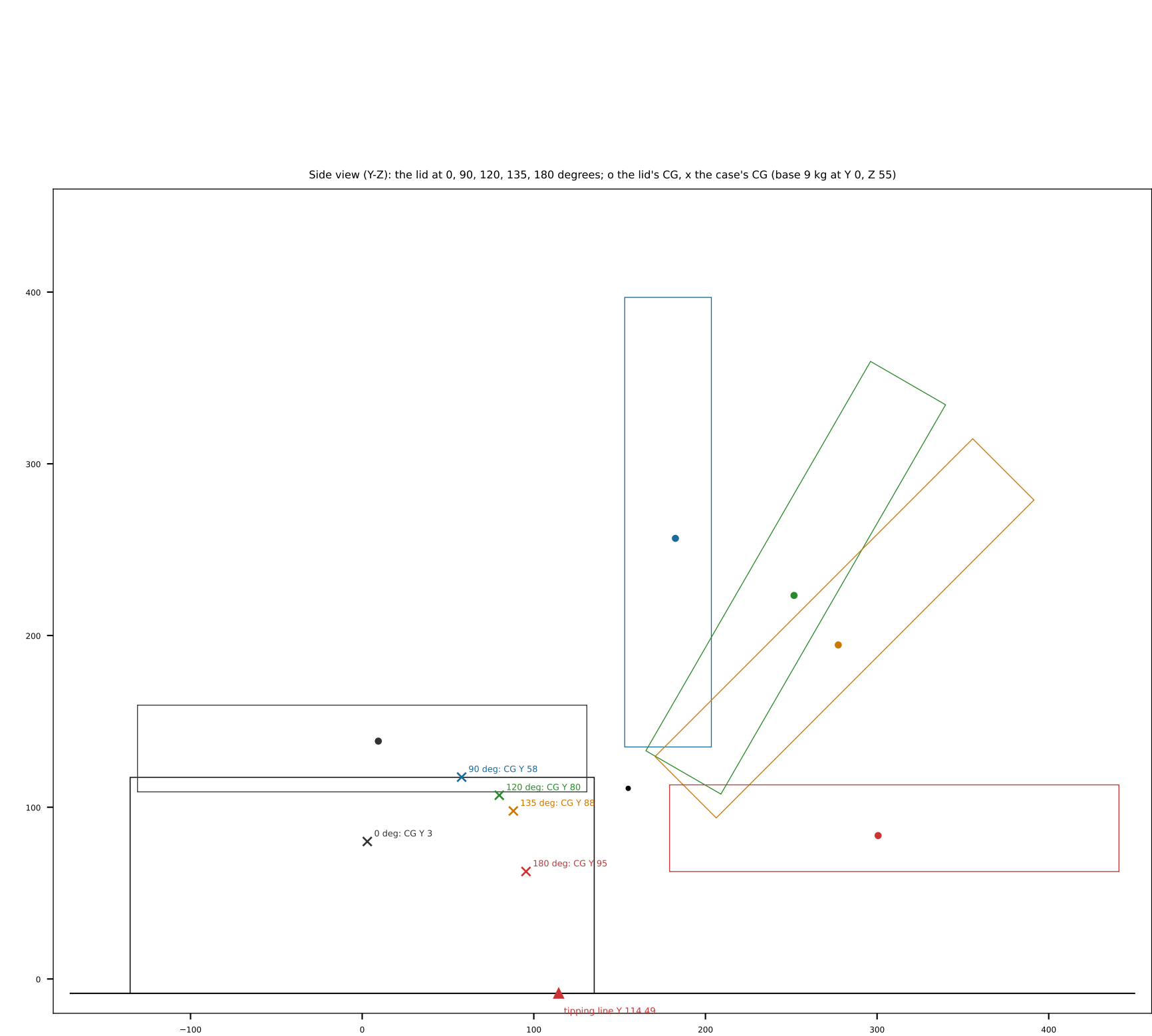
MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



CURRENT (ASSUMPTION for fnd/alelec): 10 A continuous, 18 A for 60 s, 8.6 A charge
CONDUCTOR 12 AWG fine-strand silicone, OD 4.4 (class); 2 x 0.80 m loop 8.34 mOhm:
150 mV and 2.7 W at 18 A; adiabatic rise 8.9 K over 60 s
TRAVEL the lid tie turns at R 27.0 about the axis; the chord T_L..T_P runs
150.2 closed to 161.6 at 180 degrees; lead 169.7 between the ties
BEND closed bow 33.1 (radius about 102), open S radius about 94,
against 10 x OD = 44 (the requirement on the pick)
STRAIN RELIEF bonded tie mounts at T_L, T_P and every 60 mm or less;
a printed guide keeps the lead inboard of the lid wall (Y < 130.9)
DISCONNECT XT60 pair and JST XH 1x4 beside P2 (the lid comes off)
CROSSING the sealed face: S-95 (EQ-31), NOT designed: +2 power poles
of 25 A or more and 4 signal contacts, sealed mated and unmated

MeshSat field kit V2, Option A(i) lid pack: sheet A1-5 mass and open-lid stability

MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



LID, arrangement B: 4.89 kg in all			
lid pack cells, 56 x 50 g max (spec 3.10)			2.800
lid plate 5052-H32 2.0 (2.68 g/cm3)			0.248
cover, printed V-0 class 1.0 (1.25 g/cm3, ESTIMATE)			0.105
nickel strip and group links (ESTIMATE)			0.097
wrap, fish paper, cell glue, pad, standoffs, bond (E			0.114
P2 protection board (ASSUMPTION, board P class)			0.080
lid harness to the hinge: 2 x 12 AWG 0.45 m, sense l			0.080
QMX set r2: unit and plugs 0.240, tray, frame, plate			0.372
lid shell, 39.7 percent of 2.5 kg (ESTIMATE)			0.993

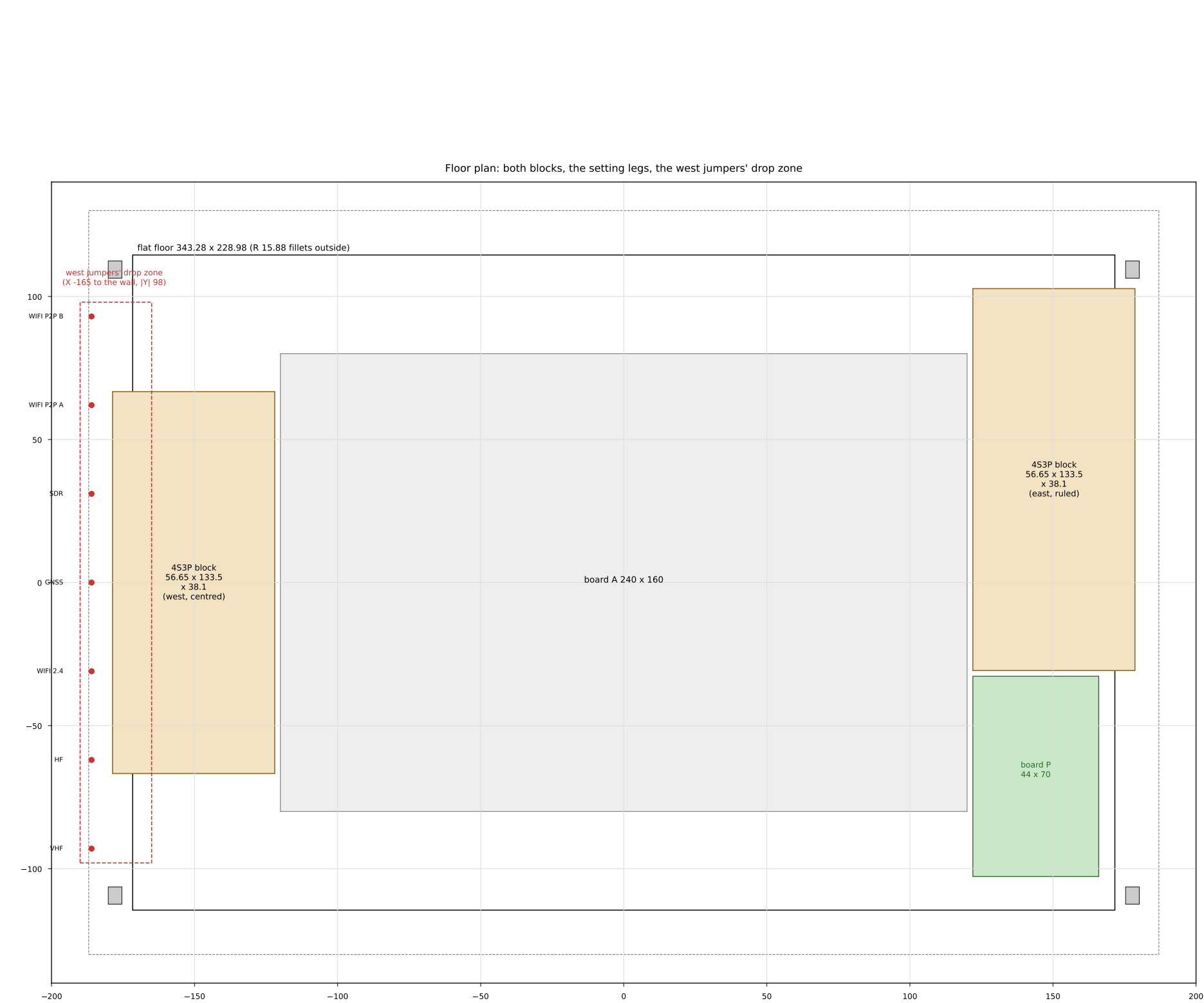
WORST OVER THE SWEEP (base 7..14 kg, CG Y -20..+20, Z 45..65, hinge axis Y 147..163): critical back slope, deg			
lid open	B	A	
90	14.9	16.0	
100	11.6	12.7	
110	8.4	9.6	
120	5.4	6.6	
135	1.4	2.6	
150	-2.1	-0.7	(tips on level ground)
180	-4.8	-2.9	(tips on level ground)

(the hinge axis at Y 155, Z 111 here; the .out sweeps Y and Z)

VERDICT: stands on level ground to 120 degrees of opening in every swept case; a light base tips from 135. FIX (SESSION, reversible): a lid stay stopping the lid at 100 degrees (webbing strap at X -179, a tab bonded to the lid's west end wall, a tab under the plate's 6-32 at Y 75.95; 56 to 63 N static; about EUR 10). With it: back slope 9.5 deg (B), 7.0 (C) at the worst of the full sweep, 3.6 (B) with the feet inboard; a backward push of 9 to 13 N at the QMX's height tips the lightest case (lid_pack_a1.out section 4). The stop angle, the hinge axis and the case's CG are read on the mock-up (T-A1-3); Peli states no load for its hinges (T-A1-4).

MeshSat field kit V2, Option A(i) lid pack: sheet A1-6 the base pockets: section 8's 4S6P at the same worst figures

MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



ROW	worst	x2	verdict
M4a	1.85	0.85	OPEN: placed by hand; T4
M4aw	1.85	0.85	OPEN: placed by hand; T4 (mirror: Peli's cavity and
M4b	7.38	6.38	MET
M4bw	7.38	6.38	MET (mirror: Peli's cavity and the RF entry plates a
M4c	3.18	1.80	MET
M4cw	3.18	1.80	MET (mirror: Peli's cavity and the RF entry plates a
M5	1.77	0.27	OPEN: placed by hand, the legs' locator; T4
M5w	37.77	36.89	MET (the block's Y place is not ruled; centred here)
M6	3.66	2.90	MET
M6w	3.99	n/a	OPEN: the x2 reading is not computed by A06's tool;

WEST RF JUMPERS: 5 of 7 west cables (Y -62, -31, 0, 31, 62) fall onto the west block; the slot over it is 7.19 at the worst less C33's zone (3.99); the only drop east of it is its 2.0 gap to board A, under one RG-316 (2.49): FAILS AS ASSUMED, held OPEN (section 8b, the M17 class): the west RF entry is re-planned before the west block is taken. Hold-down S-27 open for both blocks.

The mirror rows hold only because Peli's cavity is the same at both ends (ribs at Y 0 and +-76.2 on both end walls, R 15.88 on all four floor edges: VERIFIED) and the RF entry plates share one outline (panel1450 C4).